



Project Outlines

Course and Project Information			
Course Name	Selected Programming Language	Course Number	0602312
Year	2025/2026	Semester	☐ First ☒ Second ☐ Summer
Credit Hours	3	Project Type	☒ Practical ☐ Theoretical ☐ Combined
Number of project Phases	3 phases	Project Work Format	☒ Individual ☐ Group

Project Objectives		
No.	CLOs	Key Skill
01	Understand basic concepts of integers and floating in Python, state basic concepts of strings and lists in Python, and recognize basic concepts in lists and dictionaries.	CLO1
02	Understand basic concepts of tuples, sets, and Boolean in Python, recognize comparison and operators in Python, and state methods and functions in Python.	CLO2
03	Recognize the concepts of CSV files and explain the Matplotlib library and its usage.	CLO3
04	Write basic and simple programs using integers and float, write basic examples using strings and lists in Python, and solve examples using lists and dictionaries in Python.	CLO4
05	Solve basic concepts of tuples, sets, and Boolean, solve examples about comparison and operators in Python, and write several examples about methods and functions in Python.	CLO5



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06	Write and show examples about CSV files and use Matplotlib tool in Python.	CLO6
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Main Terminologies (optional)

Term	Definition

Tools to Use

Purpose	Tool Options
Python (<i>Jupyter Notebook</i>)	

Project Theme

Project Summary	This Python assignment consists of two main parts: a single task to calculate the monthly electricity bill based on user-entered consumption using conditional statements, followed by a two-stage University Course Registration System. In the first stage, the program collects student information and provides a menu for adding, dropping, and viewing courses using lists and basic input/output. In the second stage, the system is expanded by introducing a predefined list of available courses, enforcing rules such as preventing duplicate registrations and limiting credit hours, and generating a final summary of the student's registered schedule.		
Project Tasks	Phase	Tasks:	Mark
	Phase 1	1. Objective	5



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	The purpose of this assignment is to enable students to apply the concepts of conditional statements, arithmetic operations, and file handling in Python by developing a program that calculates an electricity bill based on unit consumption and stores the results in a file. 2. Assignment Description Each student is required to design and implement a Python program that performs the following tasks: Accepts the Customer Number, Customer Name, and Units Consumed as user inputs. Calculates the total electricity bill according to the given tariff structure: For the first 50 units → JD. 0.50 per unit For the next 100 units (51–150) → JD. 0.75 per unit For the next 100 units (151–250) JD. 1.20 per unit For units above 250 → JD. 1.50 per unit Applies an additional 20% surcharge on the total calculated amount.	
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		Displays the following output on the screen: Customer Number Customer Name Units Consumed Total Bill Amount (after surcharge) Appends the Customer Number, Customer Name, and Total Bill Amount to a text file named electricity_bills.txt, ensuring that each record is stored on a separate line.	
	Phase 2	Python Project Proposal: University Course Registration System Project Overview This project requires students to design and implement a simplified University Course Registration System using Python. The project will strengthen students' skills in: • Writing and organizing functions • Using dictionaries, lists, and sets • Designing menu-driven console applications • Implementing data validation and logical conditions	5



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	• Simulating a realistic university registration workflow Students will develop a complete system that allows adding students, managing courses, registering and dropping courses, and displaying information. Project Deliverables – Two Stages (10 points) Stage 1 Stage 1 Focus: System Design + Core Data Structures + Basic Operations Required Deliverables Students must submit: 1. Data Structures Implementation Design and initialize the following: 1.1 Students Dictionary Each student must include: • id (e.g., "S001") • name • courses → a set of enrolled course codes	
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	Sample Structure: <pre>students = { "S001": {"name": "Ahmad Ali", "courses": {"CS101", "MATH201"}}, "S002": {"name": "Sara Mahmoud", "courses": {"ENG150"}}, "S003": {"name": "Omar Hussein", "courses": {"CS101", "ENG150", "PHYS110"}}, "S004": {"name": "Lina Khaled", "courses": set()}, "S005": {"name": "Yousef Nasser", "courses": {"CS101"}} }</pre> 1.2 Courses Dictionary Each course must include: • code • name • capacity (maximum number of students) • students → a list of student IDs Sample Structure: <pre>courses = { "CS101": { "name": "Introduction to Programming", "capacity": 3, "students": ["S001", "S003", "S005"] }, "MATH201": { "name": "Calculus II", "capacity": 2, "students": ["S001"] }, "ENG150": { "name": "Academic Writing", "capacity": 4, "students": ["S002", "S003"] } }</pre>	
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	<pre>}, "PHYS110": { "name": "General Physics", "capacity": 2, "students": ["S003"] } }</pre> 2. Basic System Functions Students must implement the following fully working functions in Stage 1: • <code>add_student()</code> • <code>add_course()</code> • <code>display_all_students()</code> • <code>display_all_courses()</code>	
	3. Main Menu Skeleton Students must also prepare the main program structure: A menu that displays options but does not require full functionality yet. Example Menu: 1. Add Student 2. Add Course 3. Register Course 4. Drop Course 5. Show Student Courses 6. Show Course Students 7. Display All Students 8. Display All Courses 9. Exit	



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	Phase 3	Stage 2 Stage 2 Focus: Functional System + Validations + Final Program Required Deliverables Students must complete and submit: 1. Course Registration Functions Fully implemented and tested: • <code>register_course(student_id, course_code)</code> • <code>drop_course(student_id, course_code)</code> Features required: • Check course capacity • Prevent duplicate registration • Ensure student and course exist • Ensure student is enrolled before dropping 2. Display Functions Students must complete the following functions: • <code>show_student_courses(student_id)</code> • <code>show_course_students(course_code)</code> Output must be clear and well formatted.	5



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		3. Final Main Menu The full menu must be functional: • The program runs in a loop • User input calls the correct functions • Program exits only when the user selects Exit	
		4. Final Testing Students must: • Test all functions • Ensure no errors occur • Validate input (empty input, wrong IDs, full courses, etc.)	

Deliverables

No.	Deliverable	Standard	Format
	Jupyter (.ipynb), PDF with screenshots		

Learning Resources

Textbook(s)	Python Crash Course' by Eric Matthews
References	
Recommended Readings	
Online / Electronic Resources	https://www.w3schools.com/python/

Submission Details



جامعة العلوم الإسلامية العالمية
The World Islamic Sciences and Education University
كلية تكنولوجيا المعلومات
Faculty of Information Technology



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File Naming Format	phasex_StudentNumber (e.g, phase1_3230601111)	
	phasex_GroupNumber (e.g, phase1_G01)	
Submission Platform	☒ Moodle	
	☐ MS Teams	
Submission Deadline	Phase 01	Date: 10 – 4 - 2026 Time: 12:00 AM
	Phase 02	Date: 10 – 5 - 2026 Time: 12:00 AM
	Phase 03	Date: 25 – 5 - 2026 Time: 12:00 AM